

Green Innovation Park

Renovation Project

Group V.

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LUND
UNIVERSITY

LTH
FACULTY OF
ENGINEERING

Location of the Building

- ❑ Slottsvägen 1, 234 56 Alnarp, Sweden
- ❑ Skåne County, Southern Sweden
- ❑ Situated on the campus of the Swedish University of Agricultural Sciences (SLU)
- ❑ Green Innovation Park



Objectives And Basic Information

Project Expectations

- ❑ Renovation – Administrative to Administrative
- ❑ Improve – Indoor environment, energy use, and accessibility

Our Expectations

- ❑ Making the building open and inviting
- ❑ Sustainable approach to renovation with strong reuse incentives
- ❑ Making the building look presentable

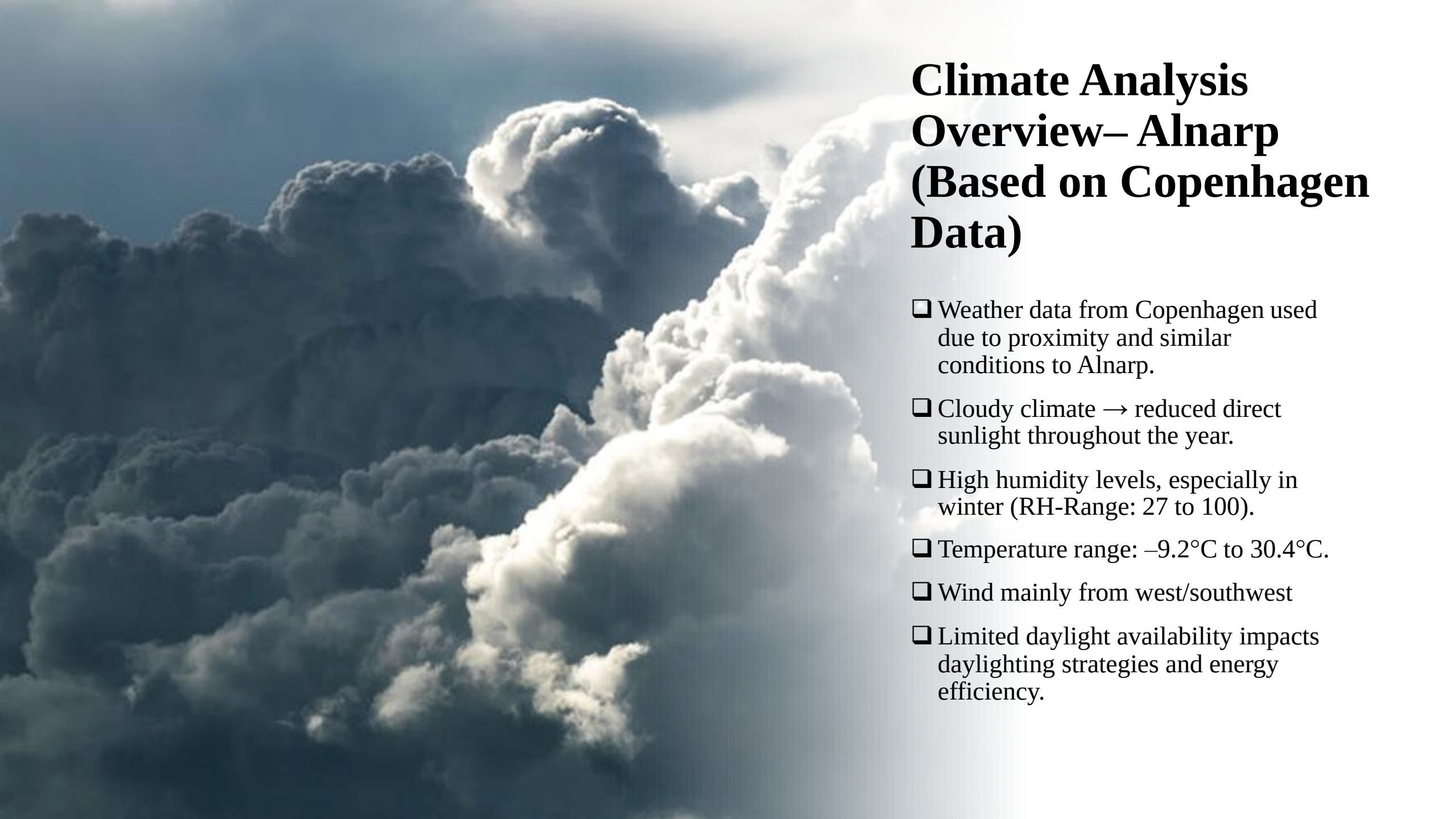
Basic Project Information

- ❑ Designed for ≈ 80 -100 people
- ❑ Heated floor area 1122 m²
- ❑ People density of ≈ 0.07 - 0.09 p/m²
- ❑ Significant emphasis on the Integrated Design Process (IDP)



Adherence to standards

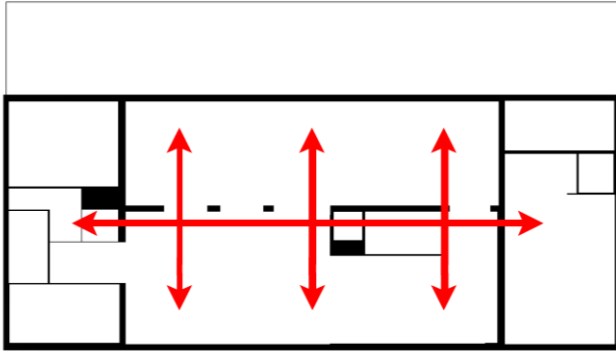
Criterion	Goal	Reference
Energy Performance	50% Reduction	BREEM 2017
Specific Fan Power	1.5 kW/(m3/s)	BBR, BFS 2011:6
Ventilation	7 l/s/person + 0,35 l/s/m2	AFS 2020:1
DGP	95% of the time with DGP below 0.35	Assignment Specific
sDA 300/50%	3 Points	LEED_v4.1
Daylight Factor	DFmedian $\geq 1\%$	BBR, BFS 2011:6
Electric Lighting	7 kWh/m2/year	EN15193-1:2017
BELOK class	26°C	BELOK
ADDITIONAL		
Primary Energy Use	<50 kWh/m2a	BBR, BFS 2011:6
UDI 300-3000	% of Fulfilment	Assignment Specific
Daylight Factor	DTM=0.7% in 95% of the room AND DT=2.1% in 50% of the room	BREEAM-SE 6.0
View Out and Sunlight Exposure	Check	EN17037:2018+A1
Daylight Factor and Illuminance	Check	EN17037:2018+A1



Climate Analysis Overview– Alnarp (Based on Copenhagen Data)

- ❑ Weather data from Copenhagen used due to proximity and similar conditions to Alnarp.
- ❑ Cloudy climate → reduced direct sunlight throughout the year.
- ❑ High humidity levels, especially in winter (RH-Range: 27 to 100).
- ❑ Temperature range: -9.2°C to 30.4°C .
- ❑ Wind mainly from west/southwest
- ❑ Limited daylight availability impacts daylighting strategies and energy efficiency.

Approach



Open Floor Plan



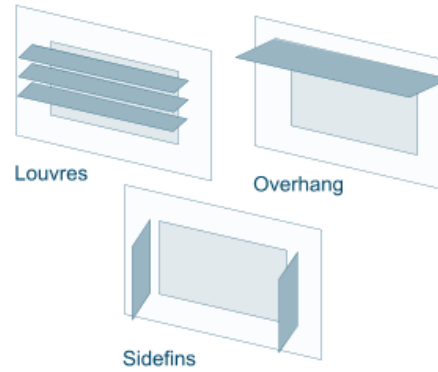
Disability Friendly



Office Environment



Natural Lighting

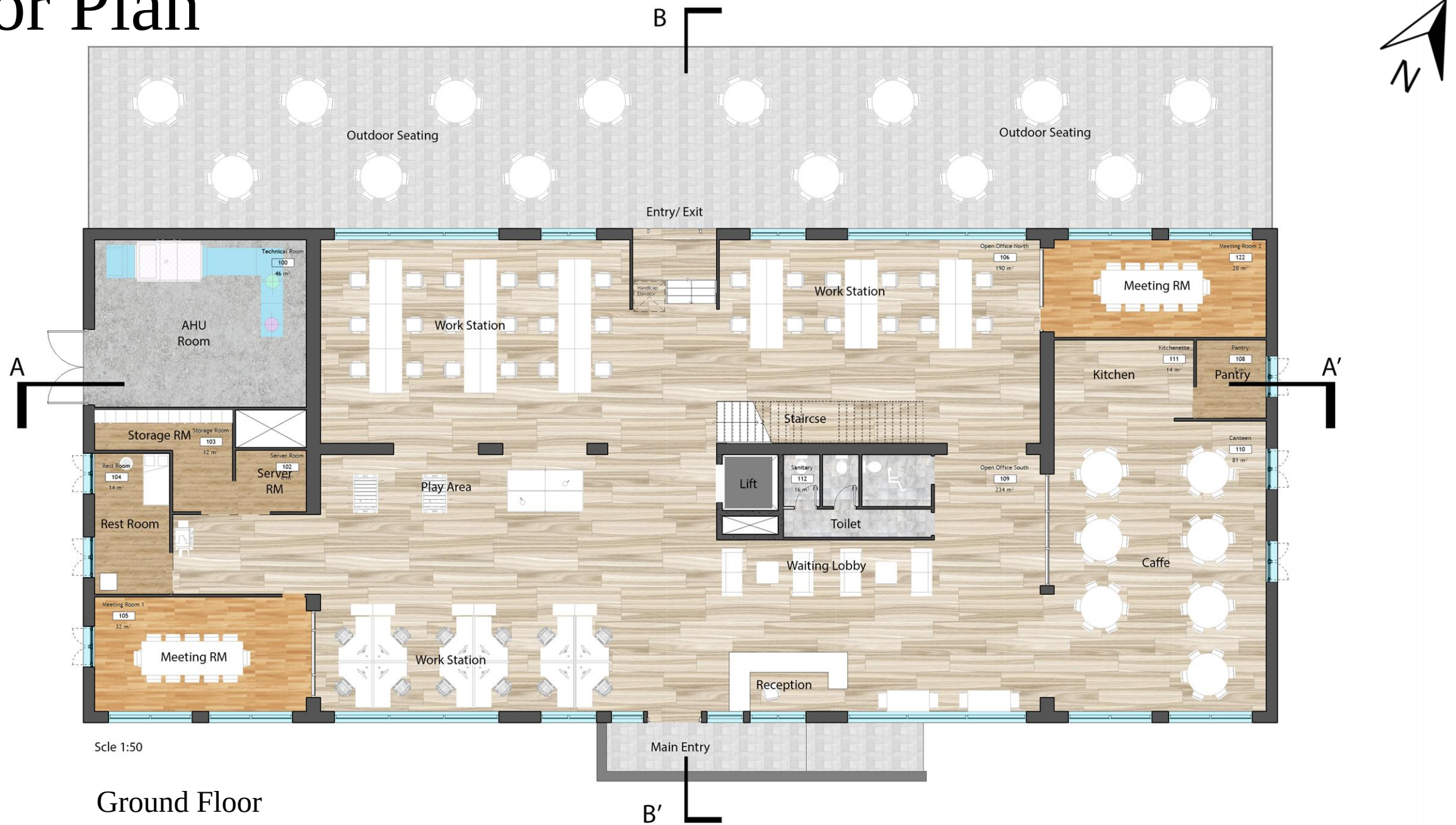


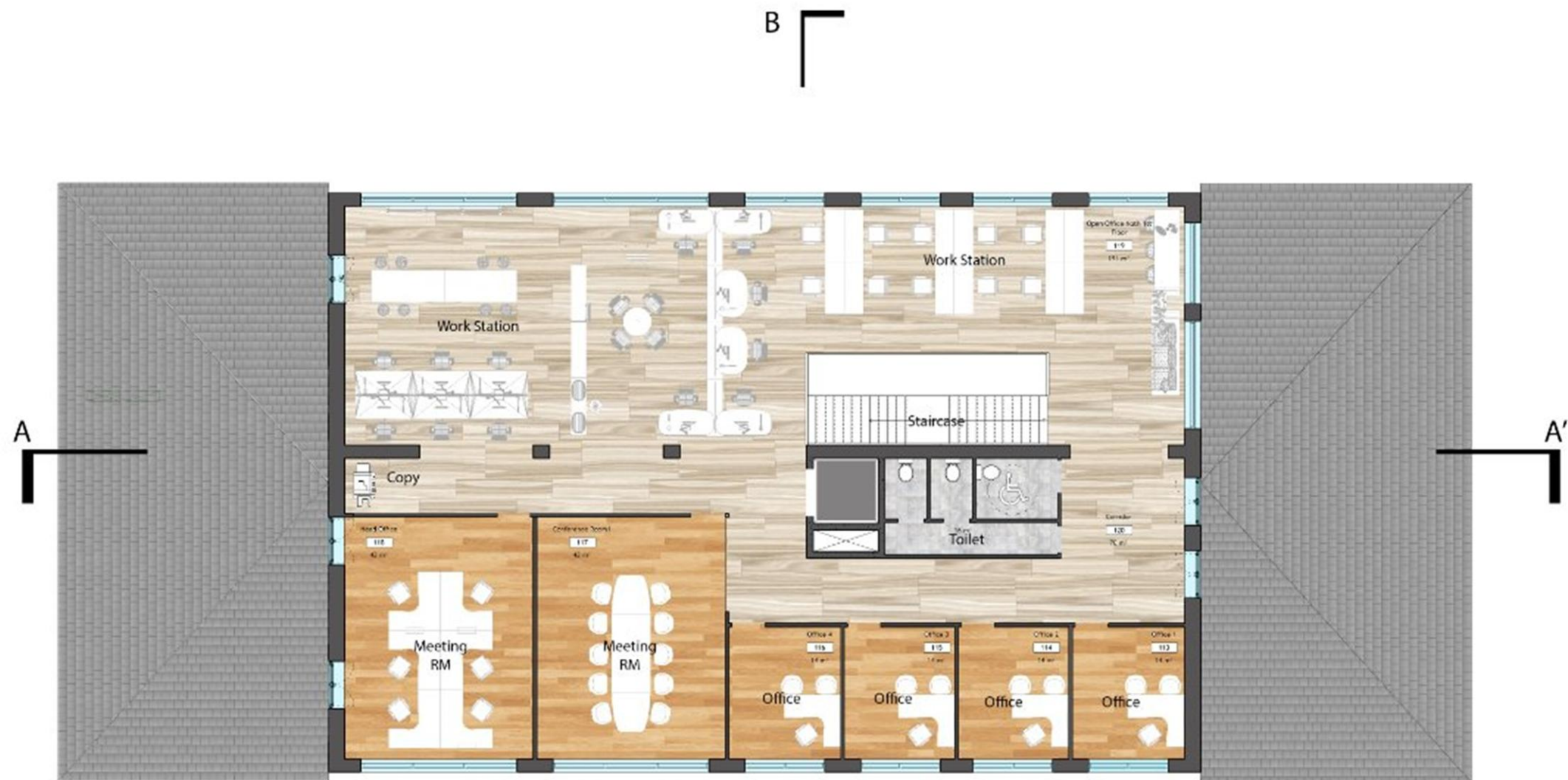
Shading Devices



Recycling Building Material

Floor Plan





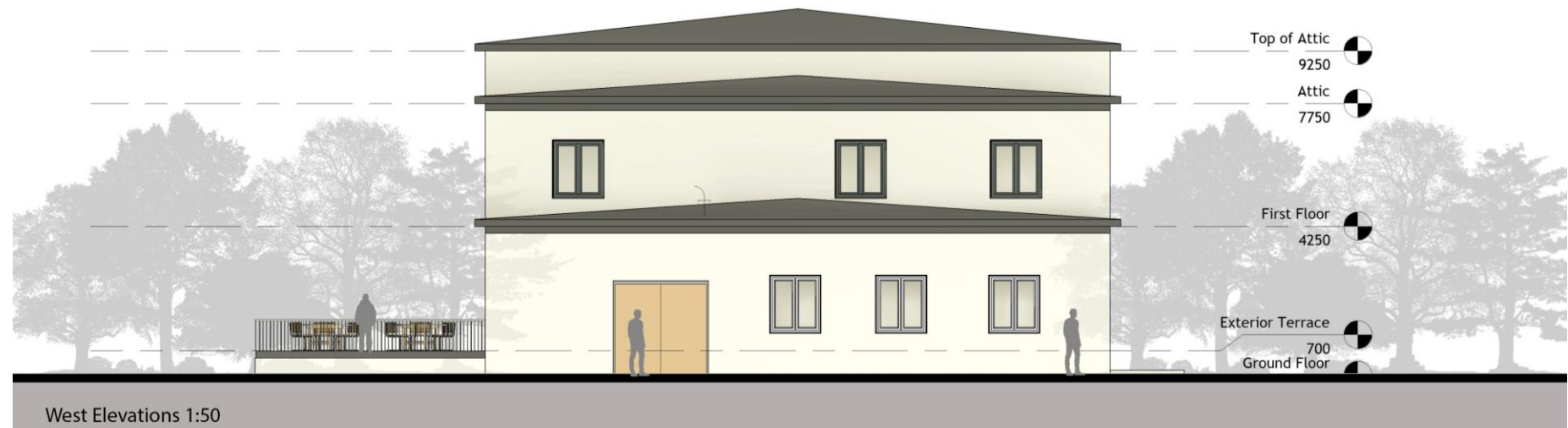
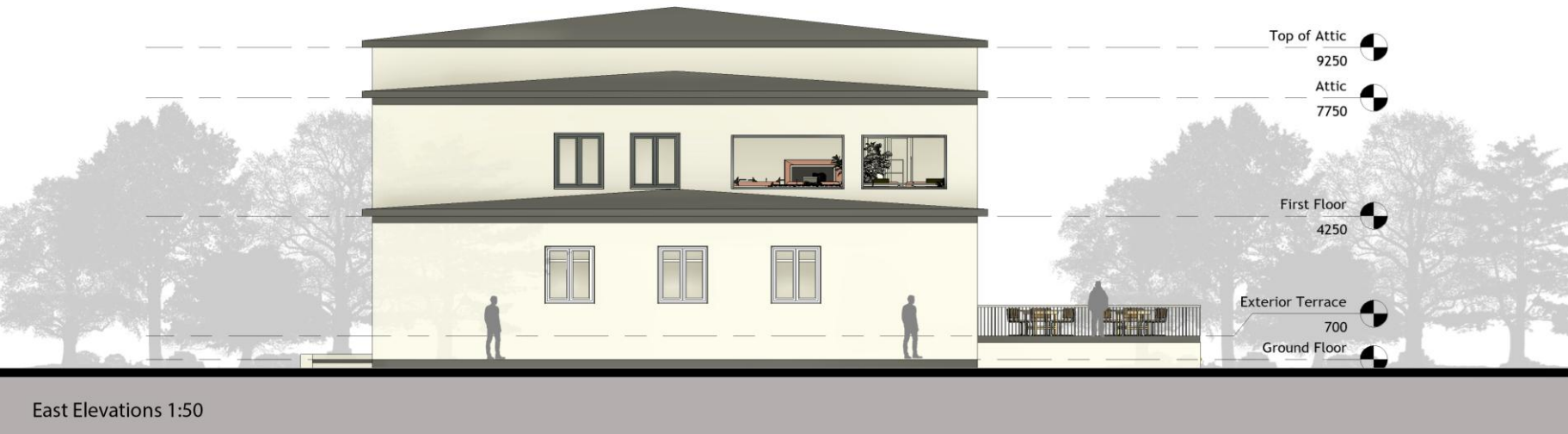
Scale 1:50

First Floor

Sections



Elevation







Daylight Performance Simulations

Daylight Factor (DF)

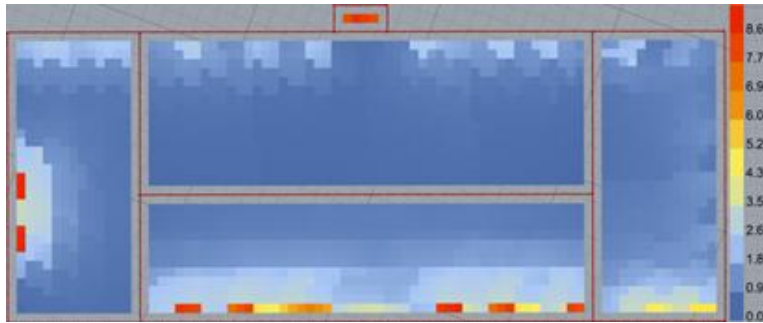
Targets To Hit

- (DF) Daylight Factor - $DF_{point} \geq 1\%$ or $DF_{median} \geq 1\%$
- (DF) Daylight Factor (BREEM-SE) - DTM=0.7% in 95% of the room and DTM=2.1% in 50% of the room

Base Case

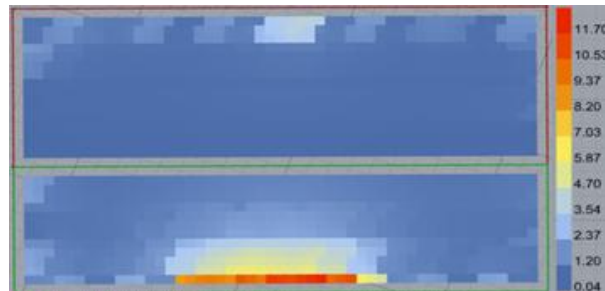
RESULTS GROUND FLOOR

DFmedian - 0.65%
DFmean - 0.98%
DTM (BREEM-SE)
- 0.7 in 45% of space
- 2.1 in 12% of space



RESULTS FIRST FLOOR

DFmedian - 0.44%
DFmean - 0.85%
DTM (BREEM-SE)
- 0.7 in 31% of space
- 2.1 in 13% of space



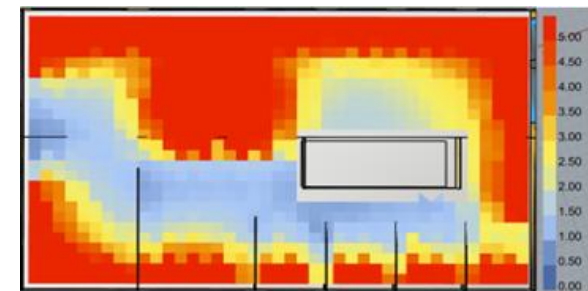
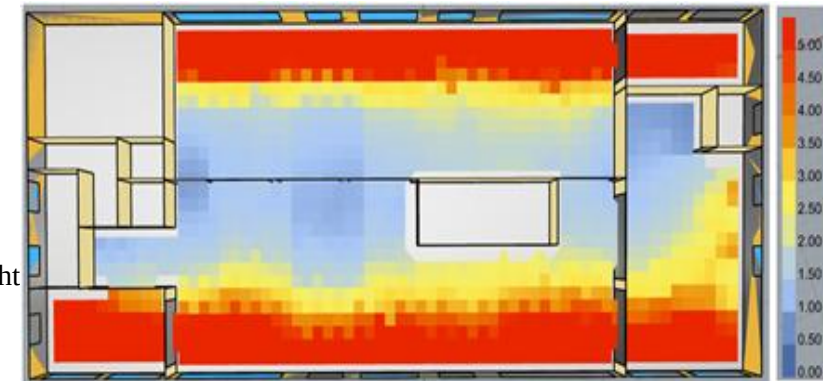
Improved Case

Enhancements:

1. Open-plan ground floor layout to improve daylight distribution.
2. Larger window openings on the north and south façades + one skylight added.
3. High-performance glazing (T-vis: 0.713, SHGC: 0.605, U-value: 0.9 W/m²·K).

RESULTS

DFmedian – 4.7%
DFmean – 2.9%
DTM (BREEM-SE)
- 0.7 in 93% of space
- 2.1 in 64% of space



Spatial Daylight Autonomy (sDA300/50%)

Targets To Hit

(sDA) Spatial Daylight Autonomy - LEED 3 points, sDA300/50% for at least 75% occupied area

Base Case

GROUND FLOOR

- 9.46%

FIRST FLOOR

- 7.98%

The building fails to meet LEED sDA
daylight autonomy requirements

Improved Case:

- 75%

Design Improvements:

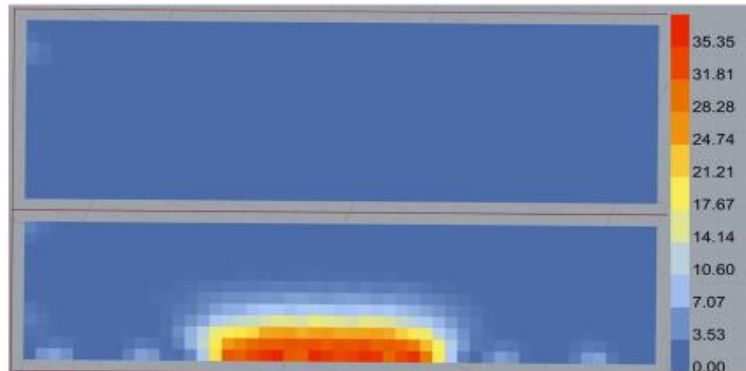
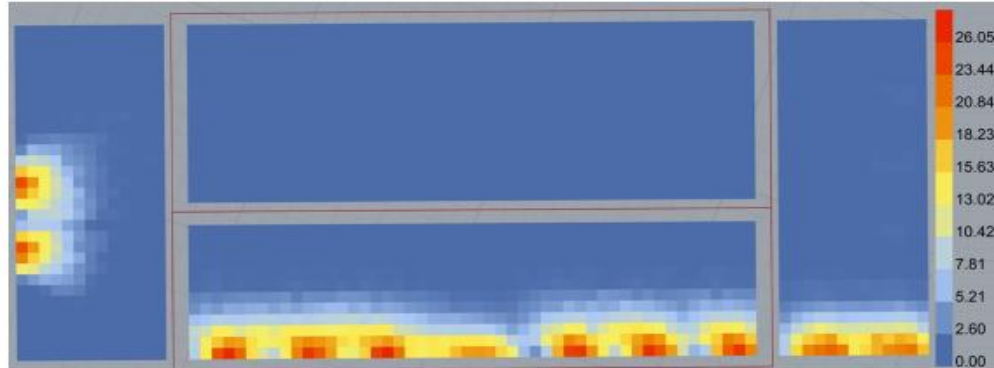
- Optimized window layout and WWR, improved glass transmission.

Annual Daylight Glare Probability – DGP - Heat-Map Based

Targets To Hit

(DGP) Annual Daylight Glare Probability - maximum of 5% of office space is affected by mostly not disturbing glare during the office hours

Base Case



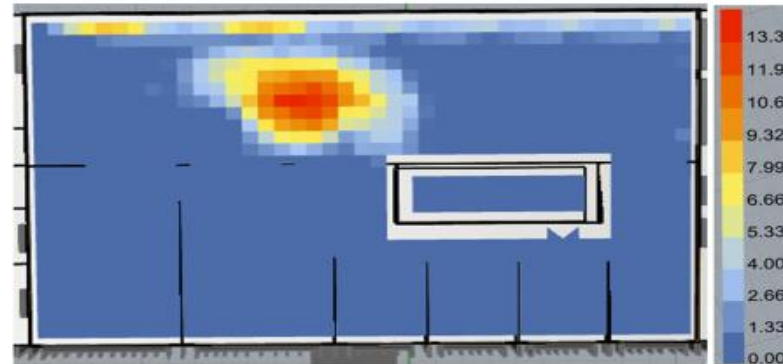
RESULTS GROUND FLOOR

- 12.85% of the office space is affected by intolerable glare

RESULTS FIRST FLOOR

- 7.21% of the office space is affected by intolerable glare

Improved Case



RESULTAT IMPROVED CASE

4.3% affected area

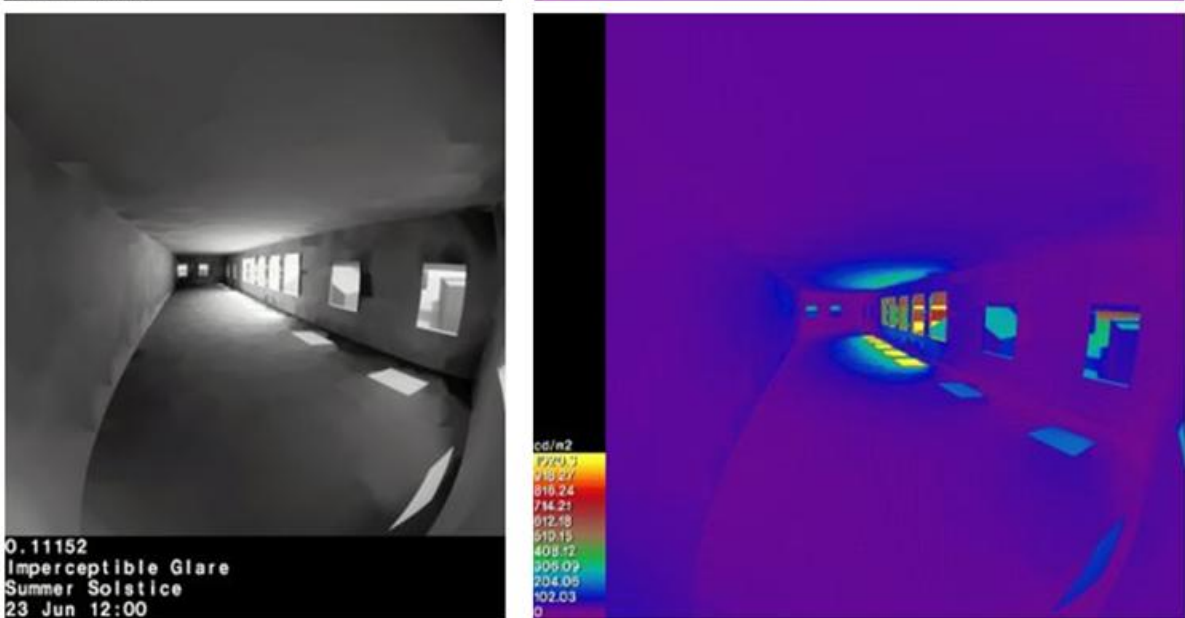
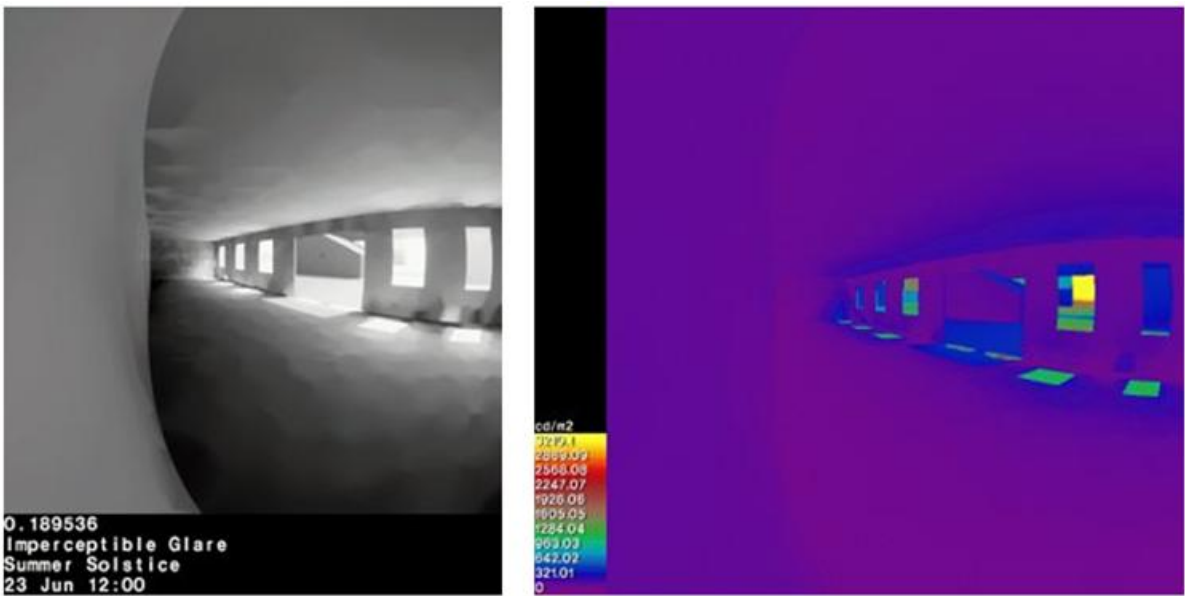
Design Improvements:

Vertical shading and dynamic blinds on south façade, dynamic blinds on east/west façades.



Daylight Glare Probability - DGP – Visual Based

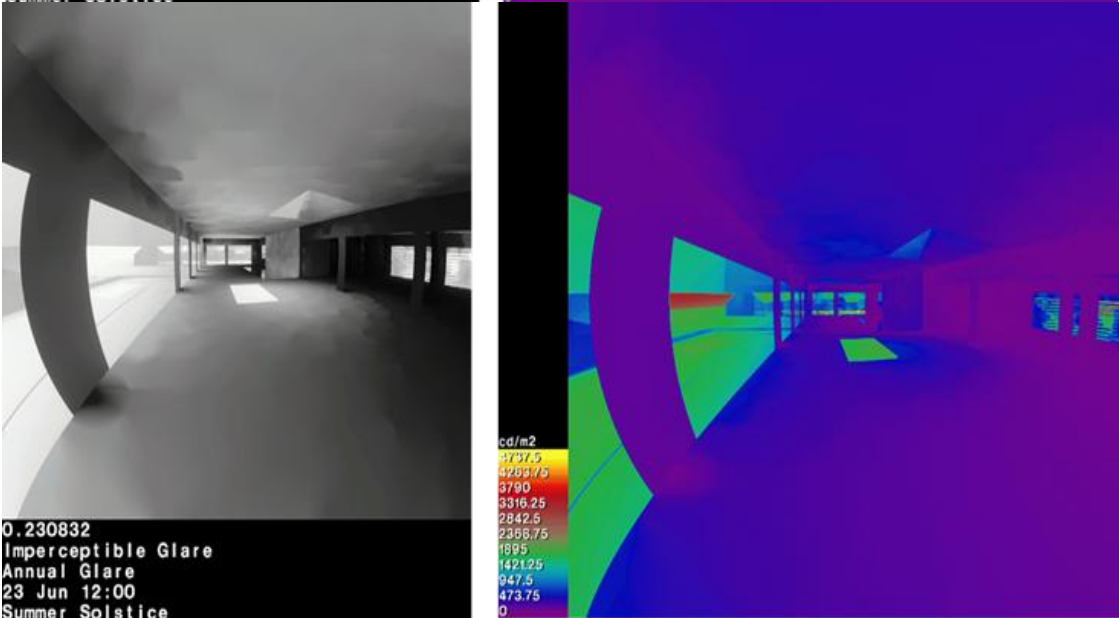
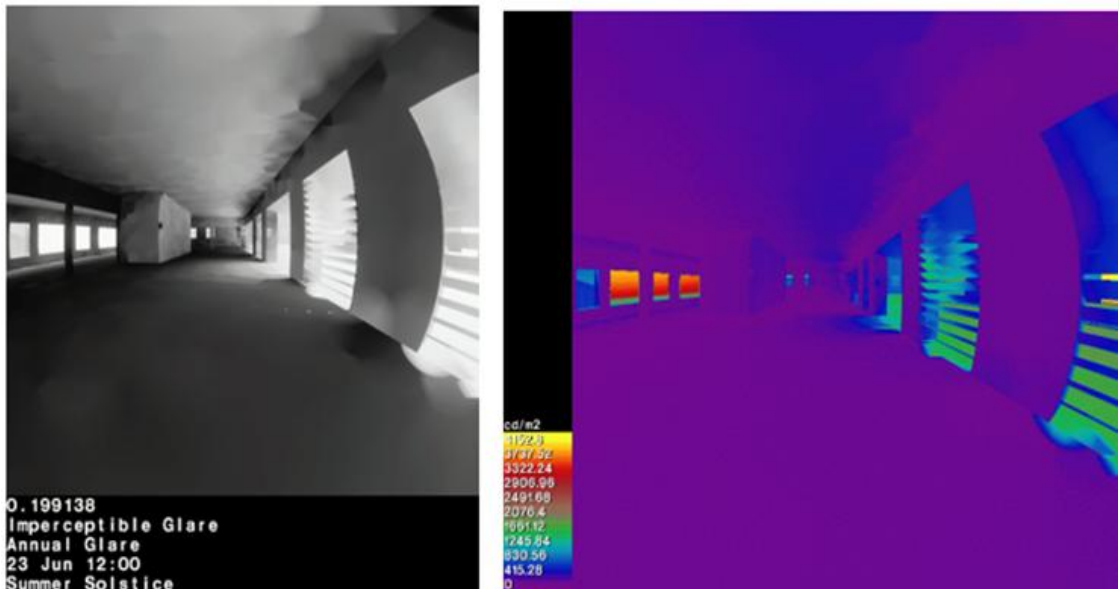
Bace Case



GROUND FLOOR

FIRST FLOOR

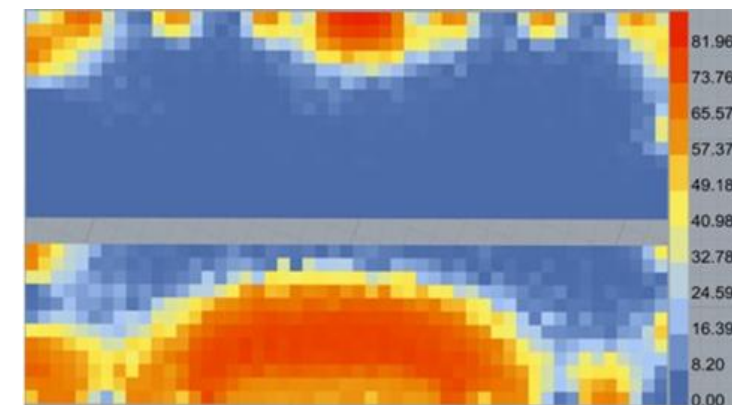
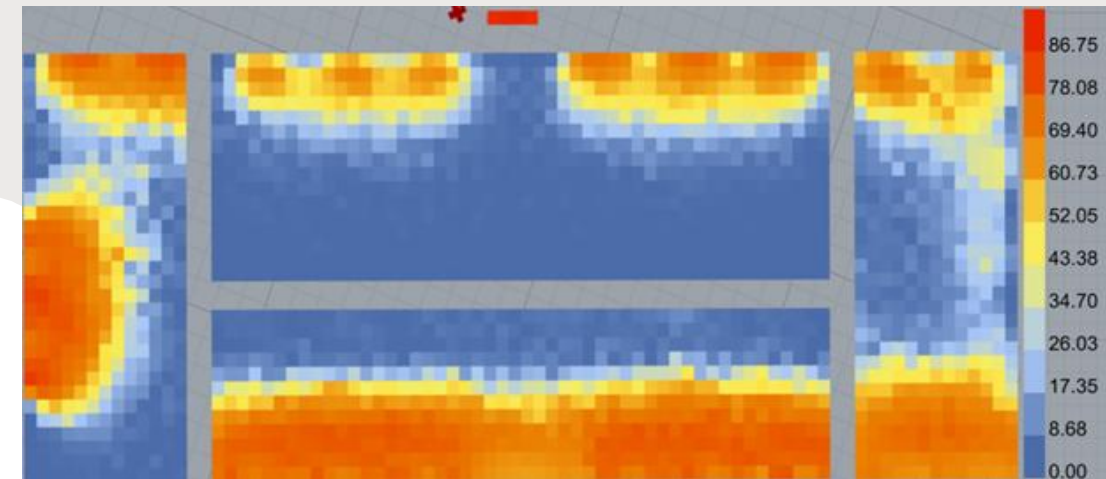
Improved Case



Useful Daylight Illuminance (UDI)



Base Case



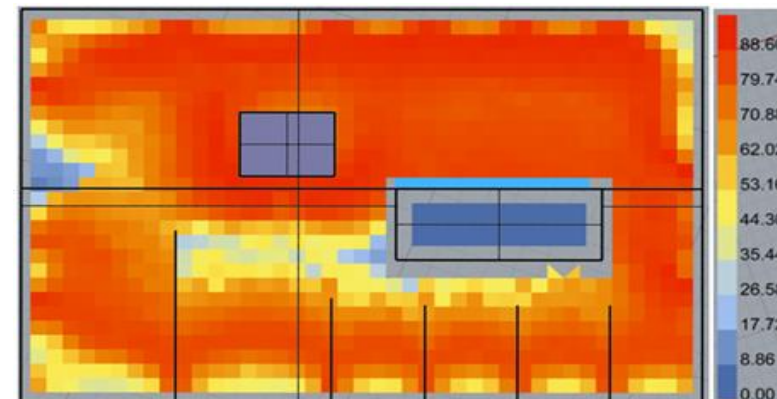
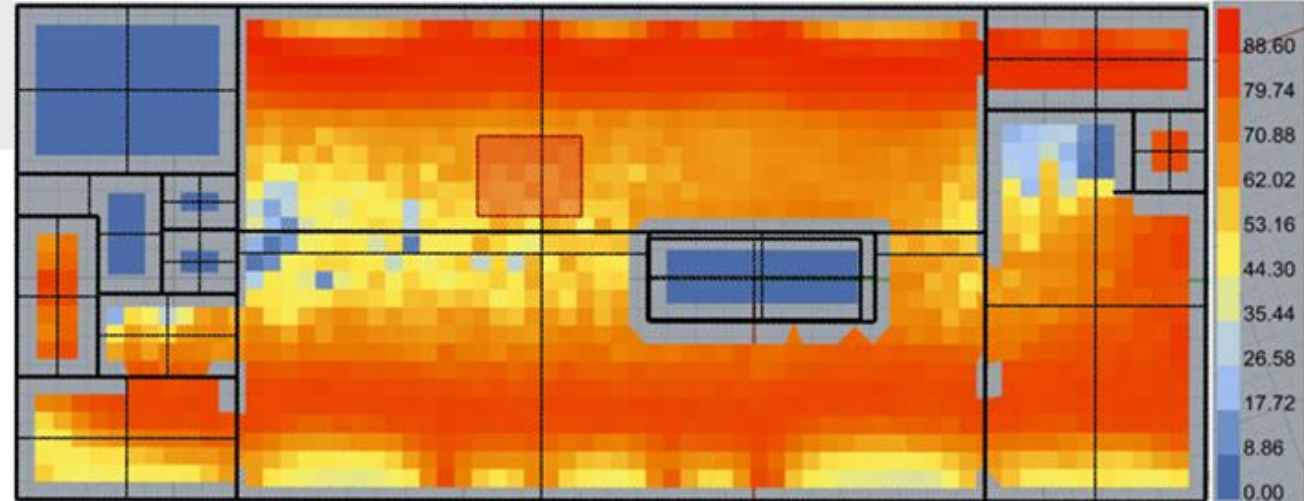
RESULTS GROUND FLOOR

- 30.3% of the space fulfills the 300-3000 lux illuminance threshold

RESULTS FIRST FLOOR

- 21.64% of the space fulfills the 300-3000 lux illuminance threshold

Improved Case



Improved Case:

- 84% % of the space fulfills the 300-3000 lux illuminance threshold

Design Improvements:

- Glazing upgrades and space reconfiguration

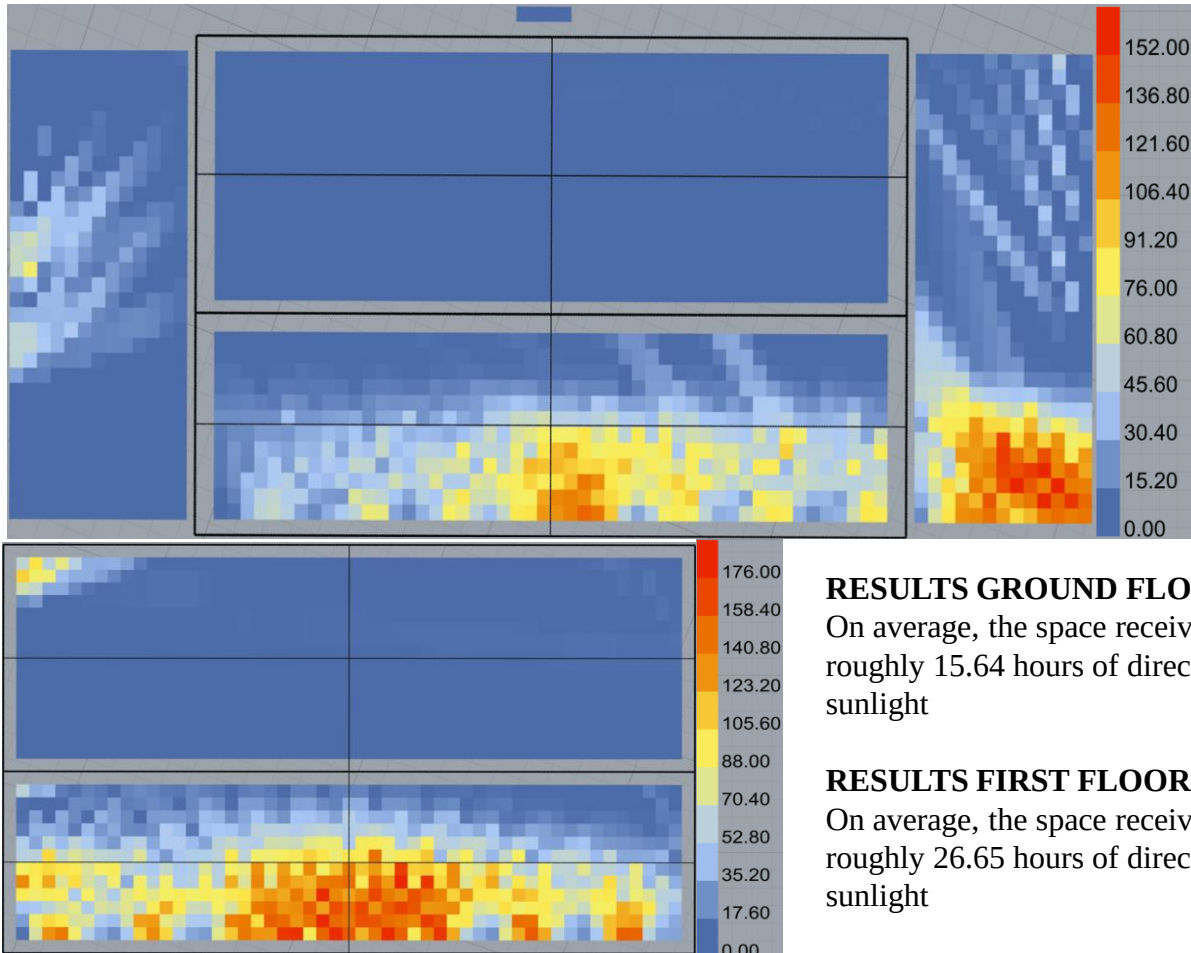
Direct Sun Exposure



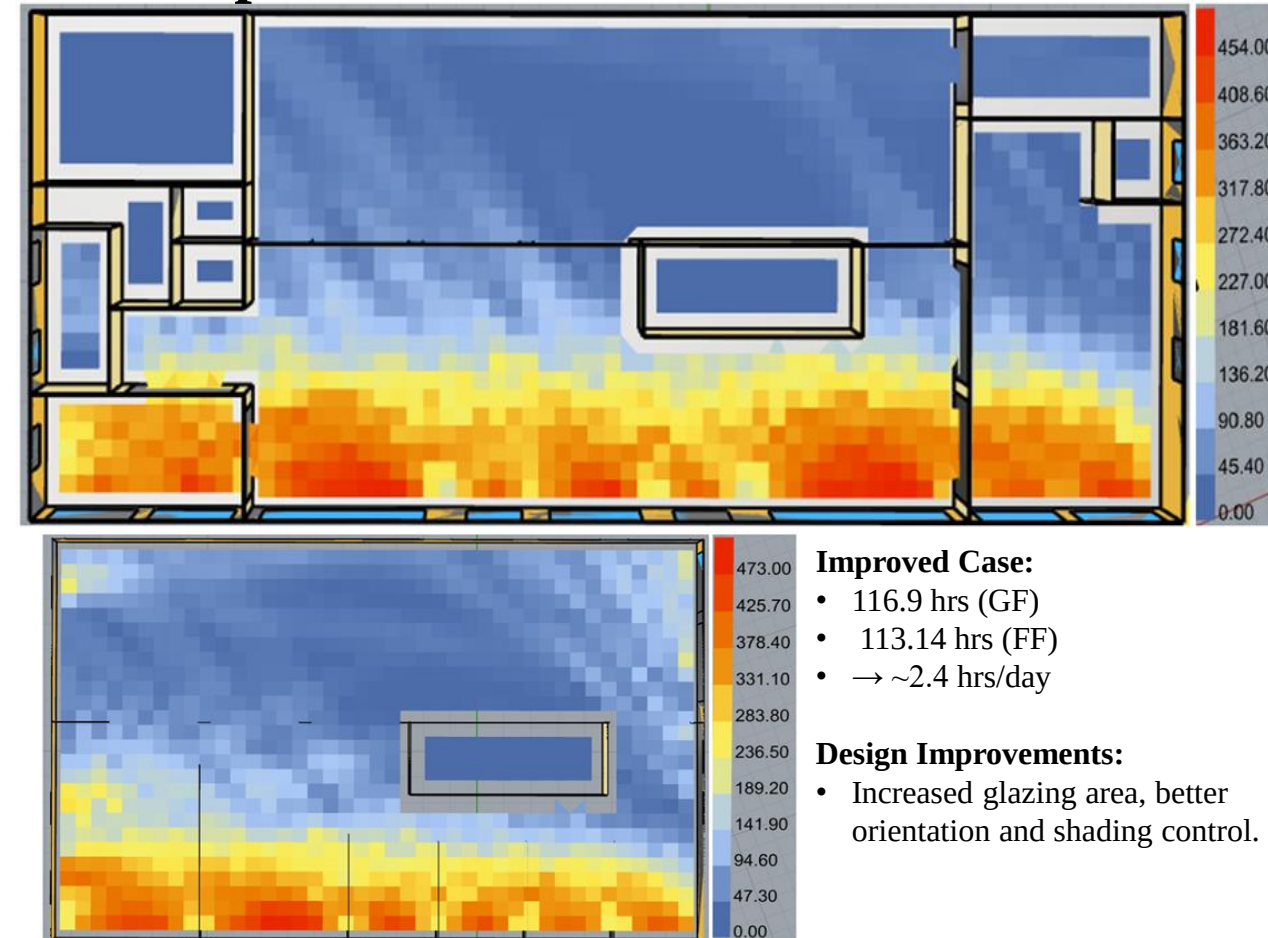
Targets To Hit

Direct Sunlight Exposure - Minimum 1.5 hours or Medium 3 hours or High 4 hours between Feb. 1st to Mar. 21st (Centre of the Glazing)

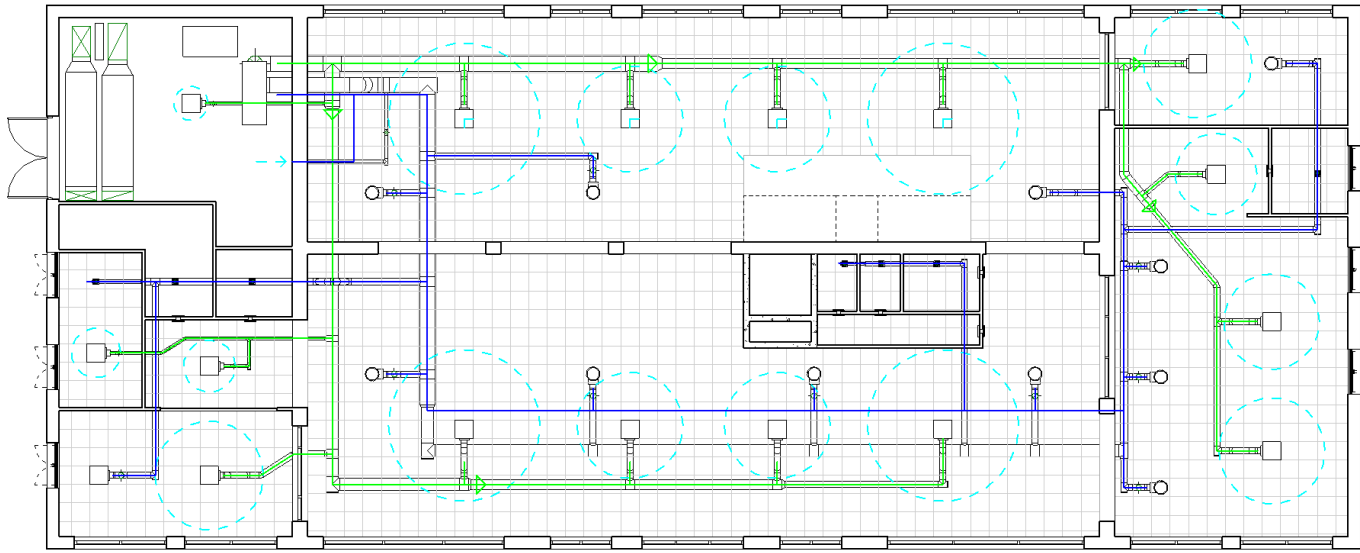
Base Case



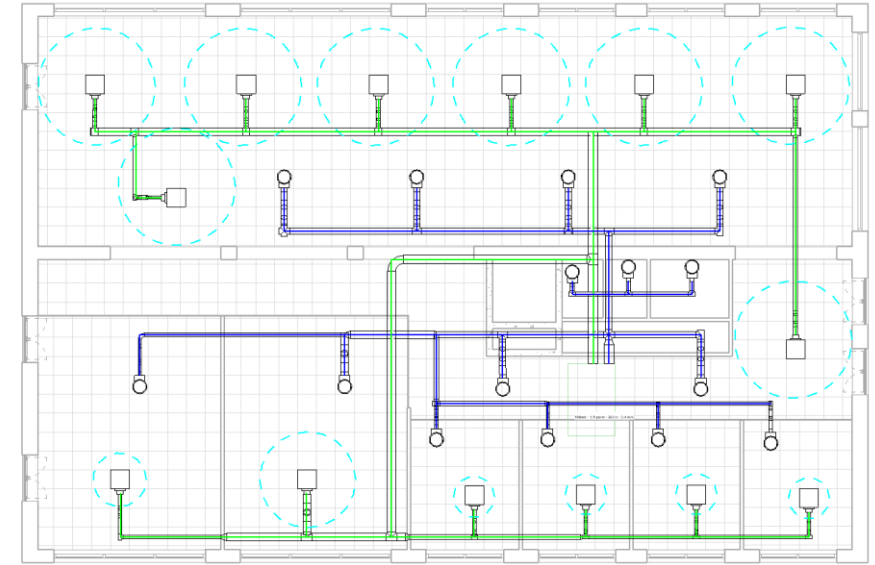
Improved Case



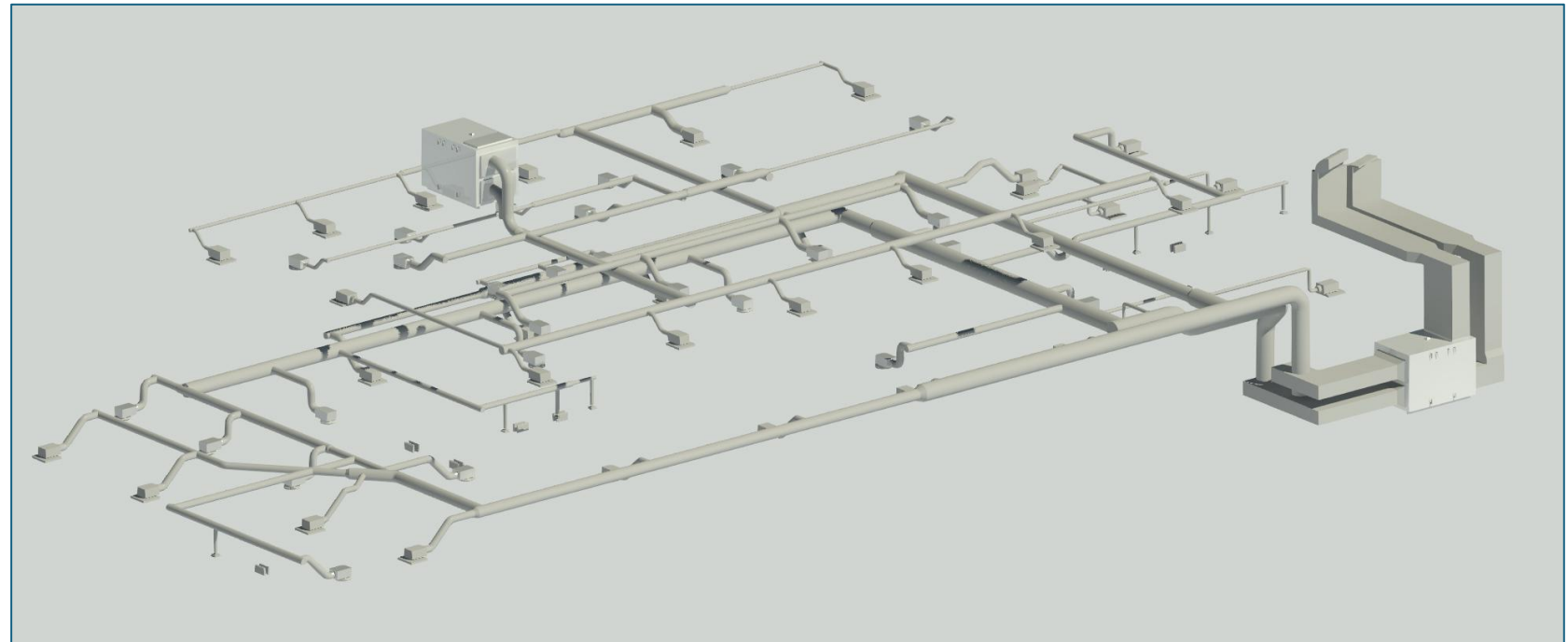
Ground Floor



First Floor



- ❑ Logical, functional and simple design approach for ducting runs
- ❑ A blue circle highlights the throw length of each supply diffuser to prevent short-circuiting between the supply and exhaust
- ❑ Where possible, exhaust diffusers are located further away from the window



Ventilation Reuse Table

DUCTING					
Diameter	Total Length	Total Reusable Length (*0.8)	Total Length In New System	Total Length of New Ducts	Reuse Percentage Out of Reusable
[mm]	[mm]	[mm]	[mm]	[mm]	[%]
100	48980	39184	55938	16754	70.0
125	128925	103140	16435	0	100.0
160	43946	35156.8	21104	0	100.0
200	54855	43884	101482	57598	43.2
250	13724	10979.2	49323	38343.8	22.3
315	11743	9394.4	35566	26171.6	26.4
400	36225	28980	37644	8664	77.0
500	21943	17554.4	29477	11922.6	59.6
800x400	4610	3688	2892	0	100.0
1000x300	9369	7495.2	9369	1873.8	80.0
1000x400	3535	2828	3535	707	80.0
1000x600	2269	1815.2	2269	453.8	80.0
1200x600	9257	7405.6	1937	0	100.0
Average Duct Reuse [%] :					78.21
AIR TERMINALS					
Total Number of Diffusers in Old System [Quantity]	Total Number of Diffusers in New System [Quantity]	Total Number of Reused Diffusers [Quantity]	Total Number of Scrapped Diffusers [Quantity]	Reuse Percentage Out of Reusable [%]	
72	65	39	33	60	

Diameter of ducts in the existing system

Total Length of ducts in the existing system

Total reusable fraction

Total length present in the system after retrofit

Total length of brand-new ducts

Percentage of reused ducts in the system after retrofit

How much can we
reuse in “**THEORY**”?

Aluminium Duct-tape



≈100%

Duct Fittings



≈??%

Air Terminals



≈60%

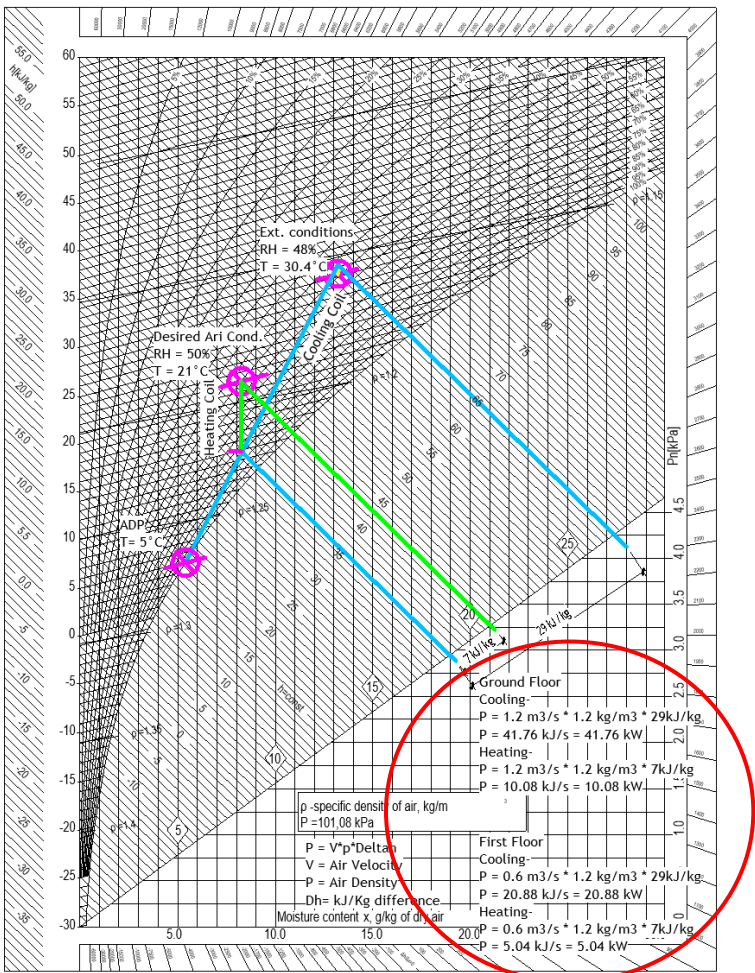
Ducting



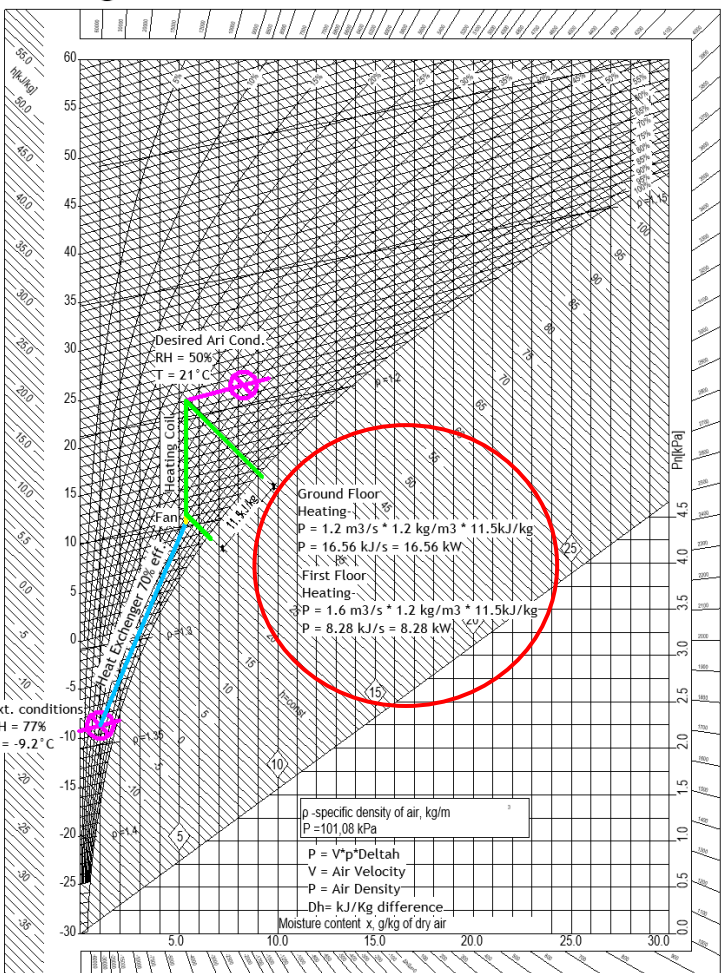
≈78%

Heating and Cooling Coils

Cooling Season



Heating Season



Ground Floor –
HC – 16.56 kW

First Floor –
Existing HC – 10 kW

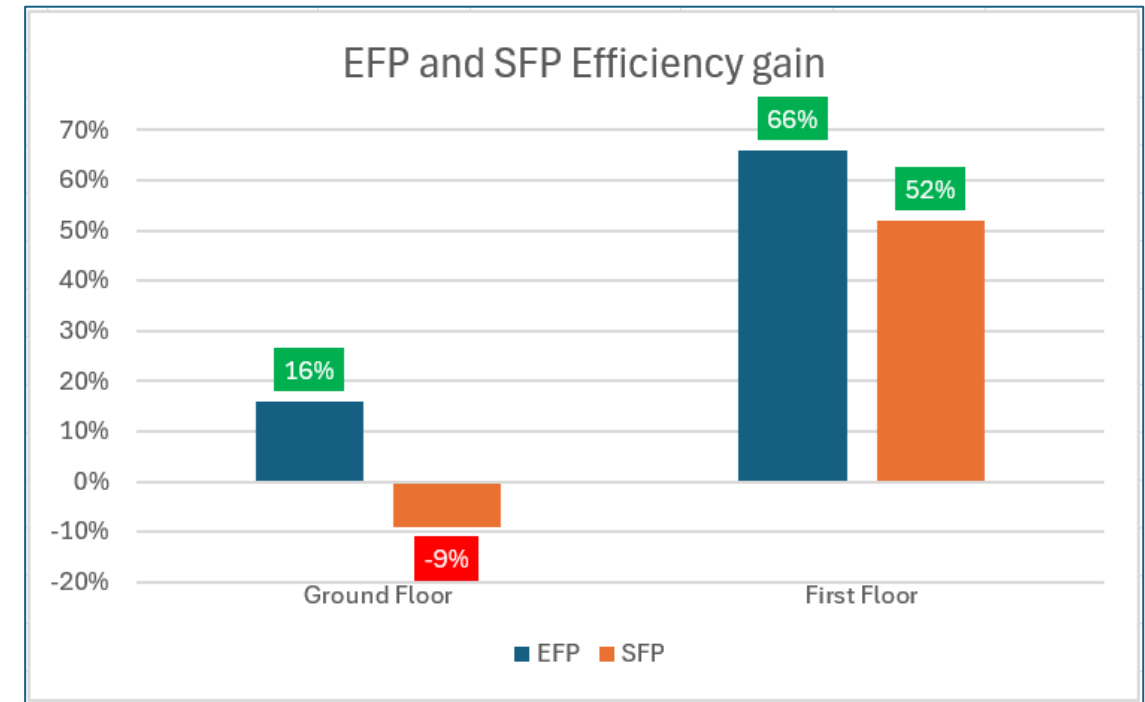
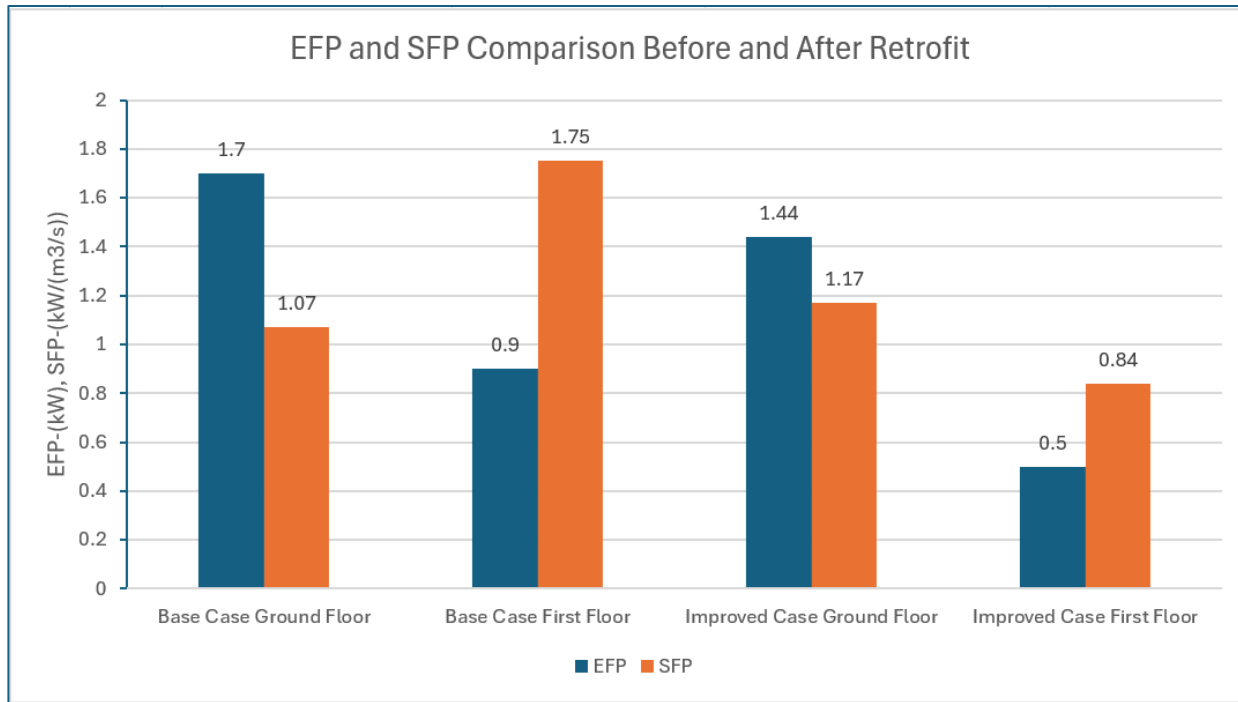


Ground Floor –
CC – 41.76 kW

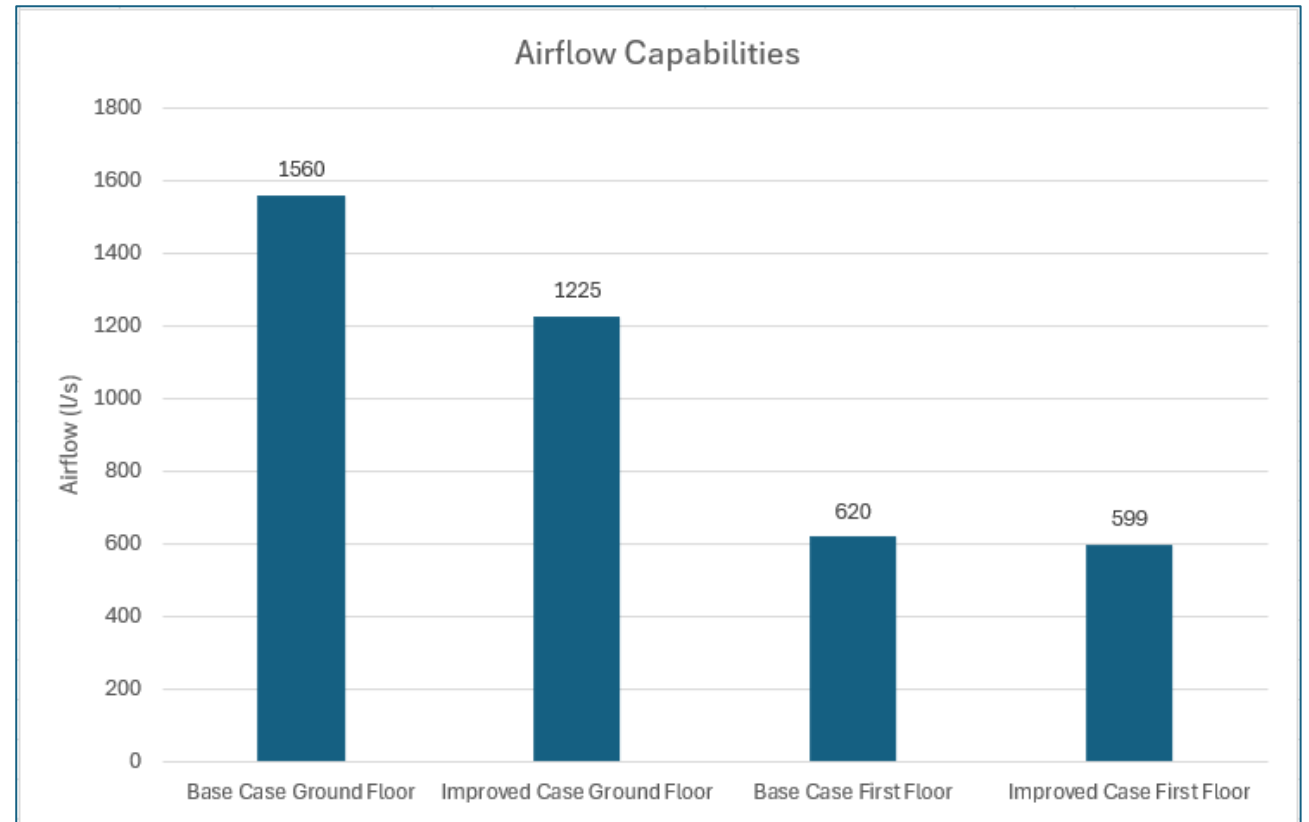
First Floor –
CC – 20.88 kW

One Existing HC $\approx$ 10kW

How does the system perform after retrofit?

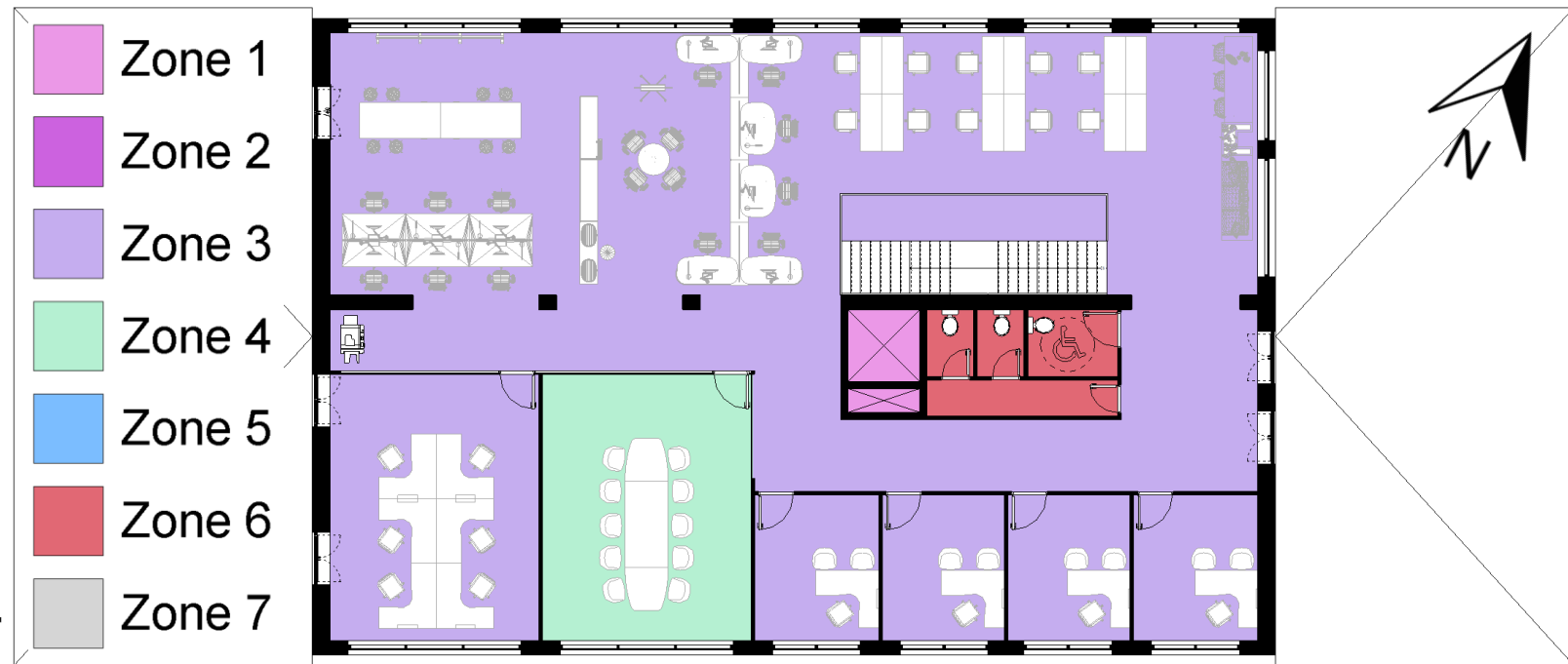


What about the
airflow comparison?

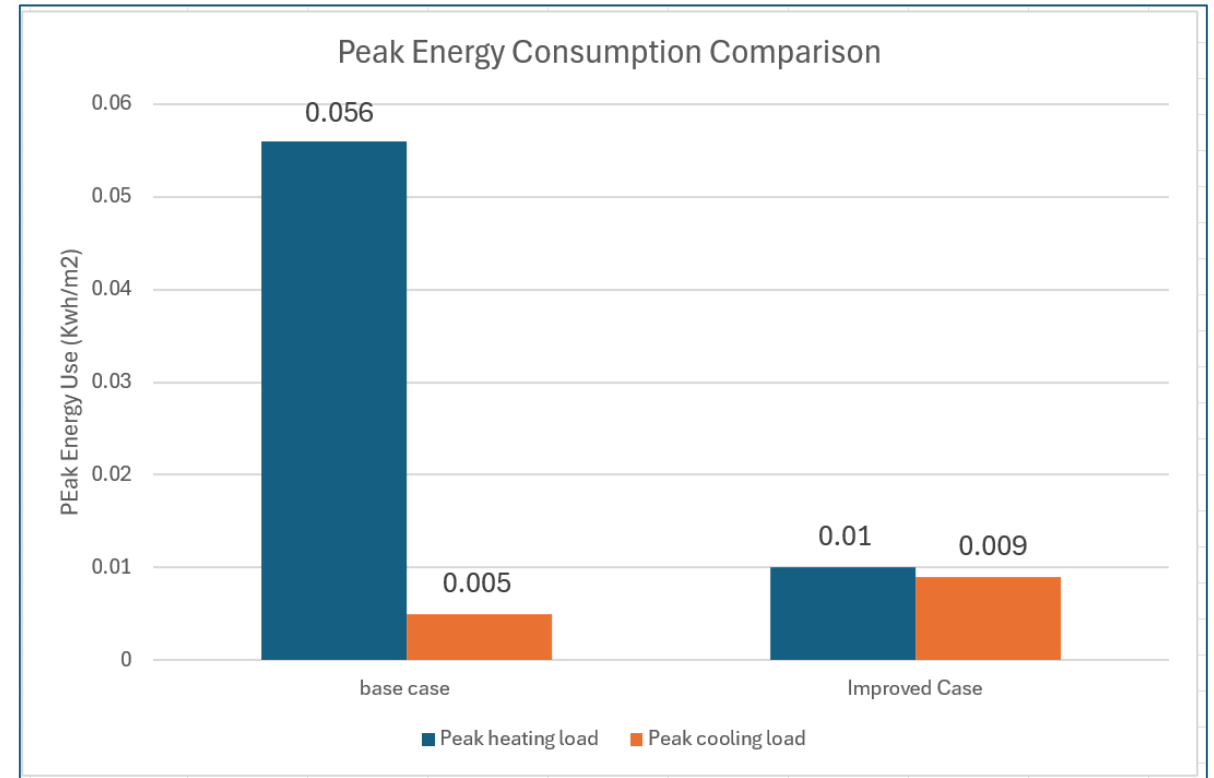


Energy Simulation Zoning

- ☐ **Zone 1** – Technical Shaft and Elevator Shaft
- ☐ **Zone 2** – Canteen
- ☐ **Zone 3** – Open Offices on Ground and First Floor
- ☐ **Zone 4** – Meeting and Conference Rooms
- ☐ **Zone 5** – Technical Room and Server Room
- ☐ **Zone 6** – Sanitary
- ☐ **Zone 7** – Pantry, Storage and Rest Room



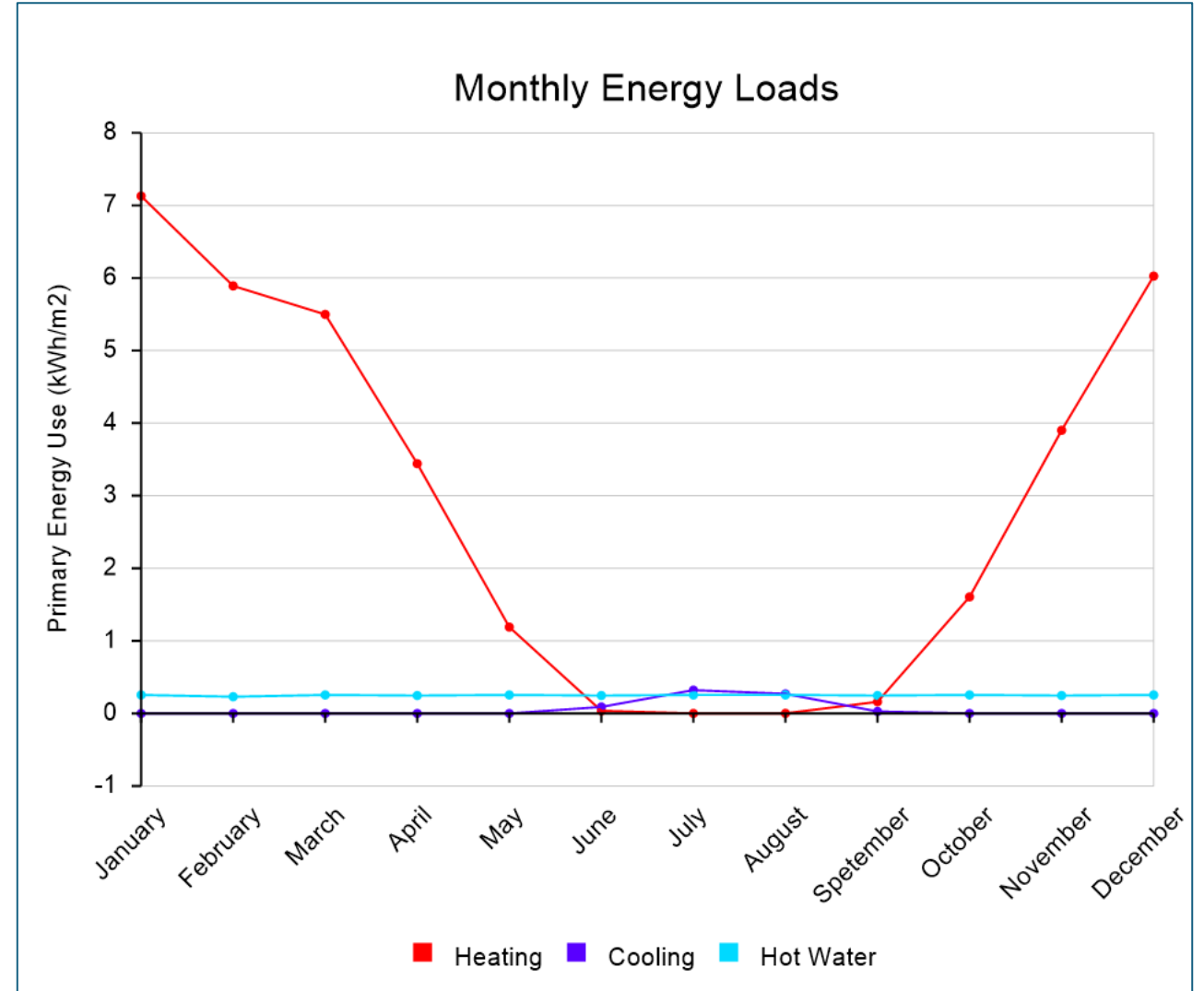
Peak Energy Consumption



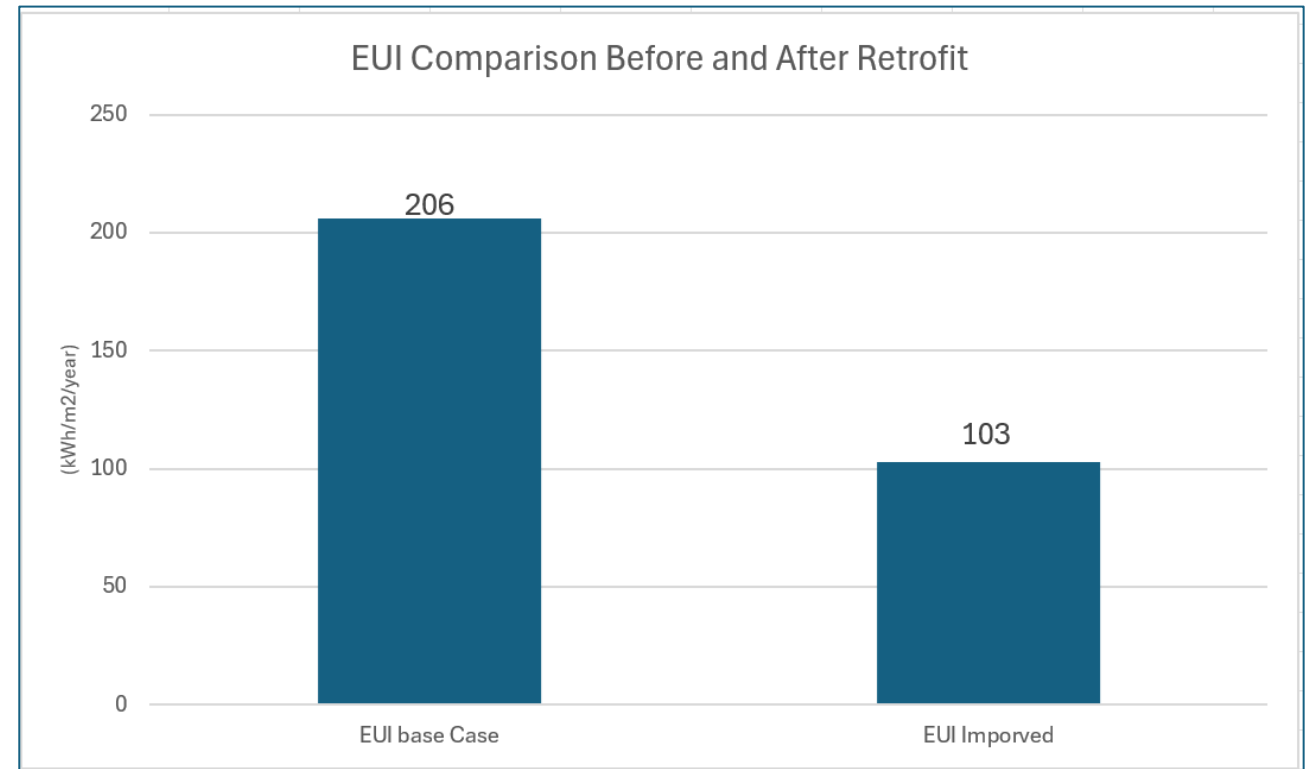
EP_{pet} Contributors

Final EP_{pet} of the project is 48.132 kWh/m²/year

Overheating during work hours between April and September is 1.1% < 10%



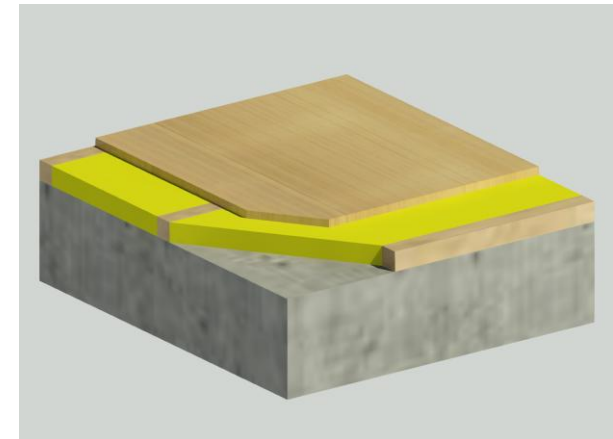
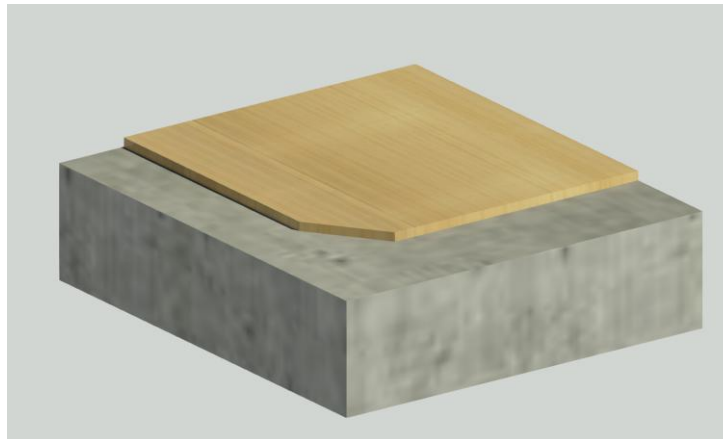
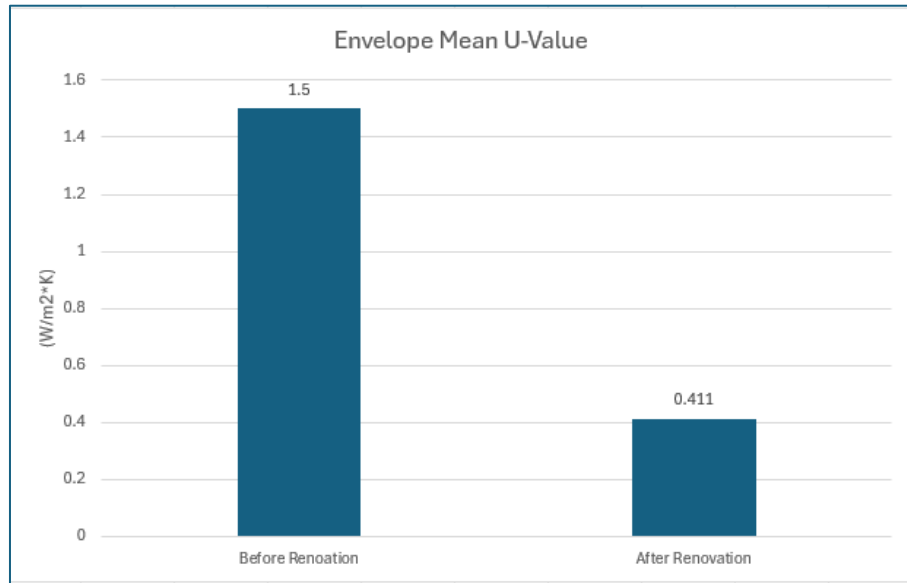
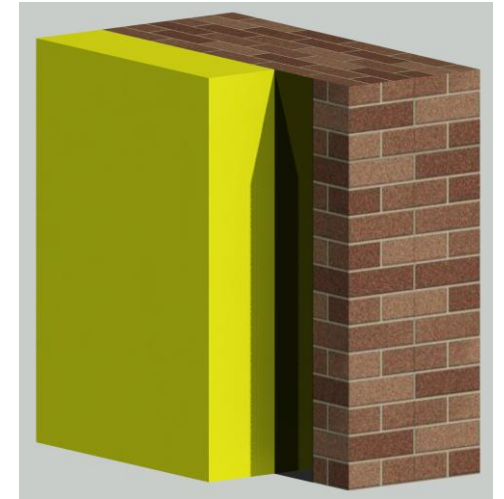
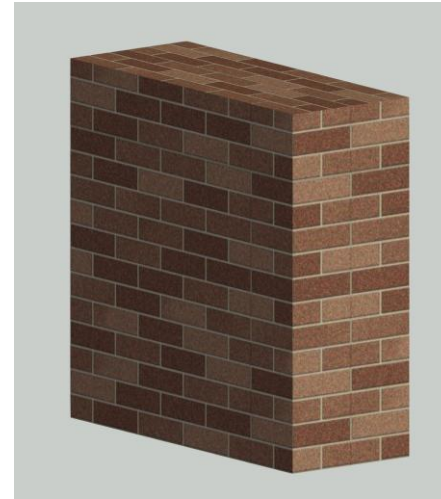
EUI Before and After Retrofit



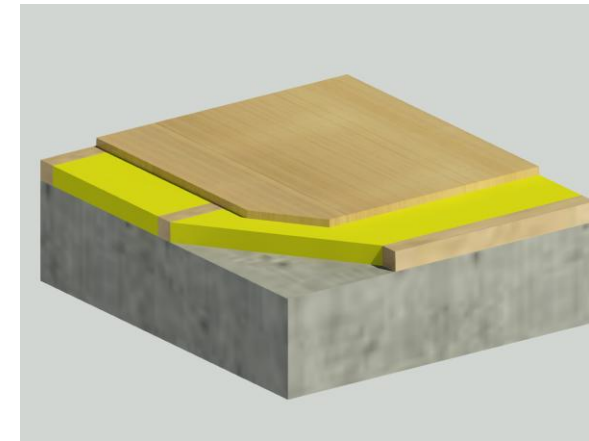
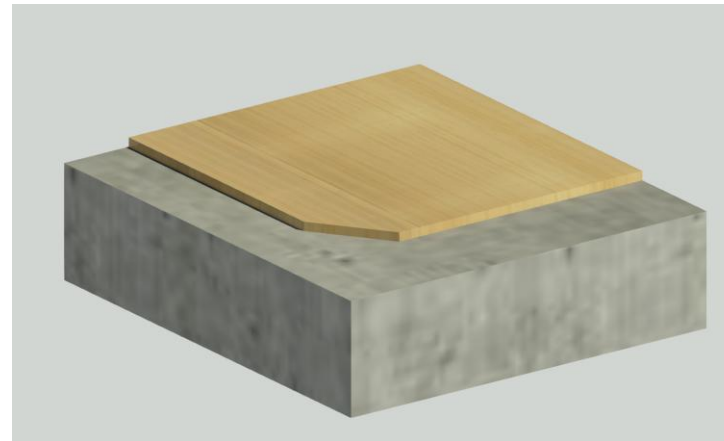
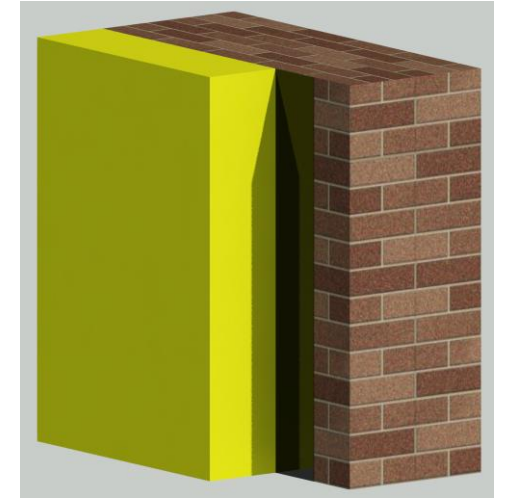
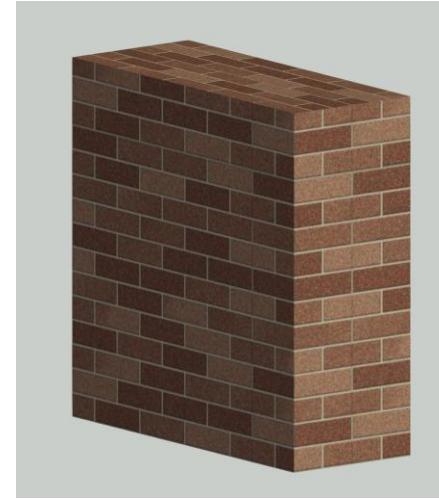
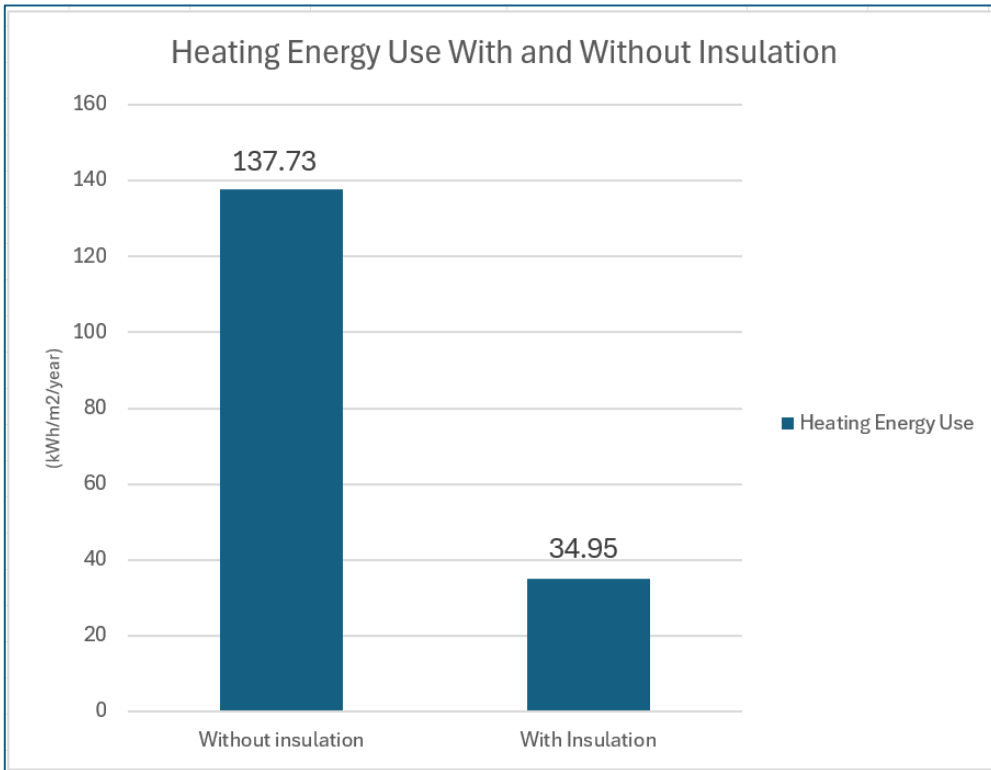
Exactly 50% reduction in Energy Use

Energy Part - Changes to Building Components

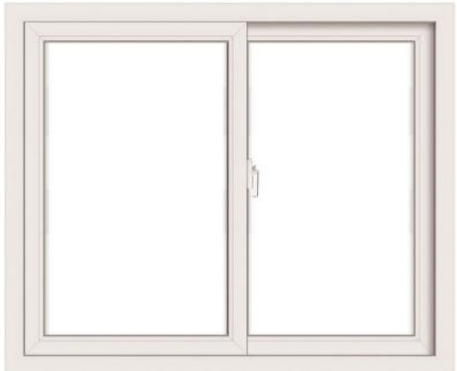
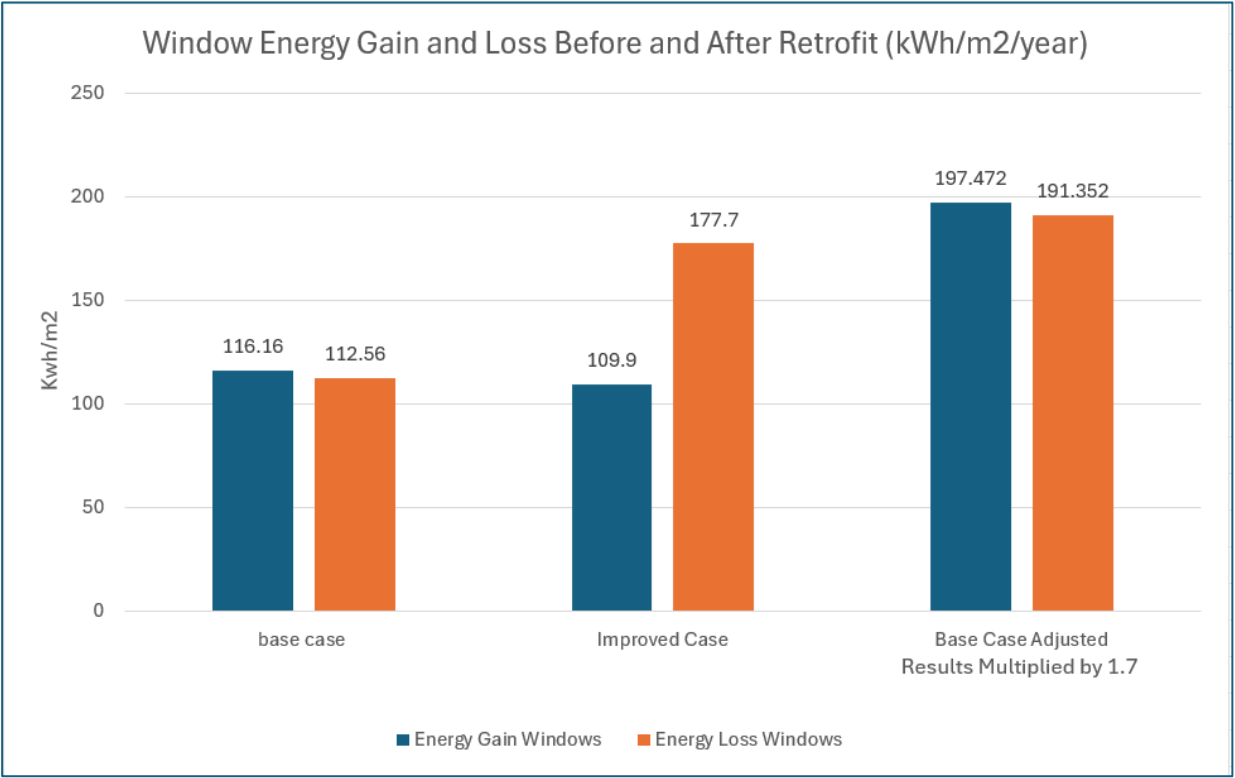
	Materials	U- Value	R-Value
	[Interior to Exterior]	[W/m ² .K]	[m ² .K/W]
Exterior Wall Base Case	Brick (400mm)	1.7	0.5
Exterior Wall After Retrofit	Brick (400mm) Mineral Wool (200mm)	0.2	6.5
Ground Floor Base Case	Concrete (400mm)	3.0	0.2
Ground Floor Wall After Retrofit	Mineral Wool (50mm) Concrete (400mm)	0.5	1.8
Roof	Concrete (200mm) Cellulose (100mm)	0.3	2.8



Heating Energy Demand with and Without Insulation Comparison



Window Performance Comparison Before and After Retrofit



New Windows

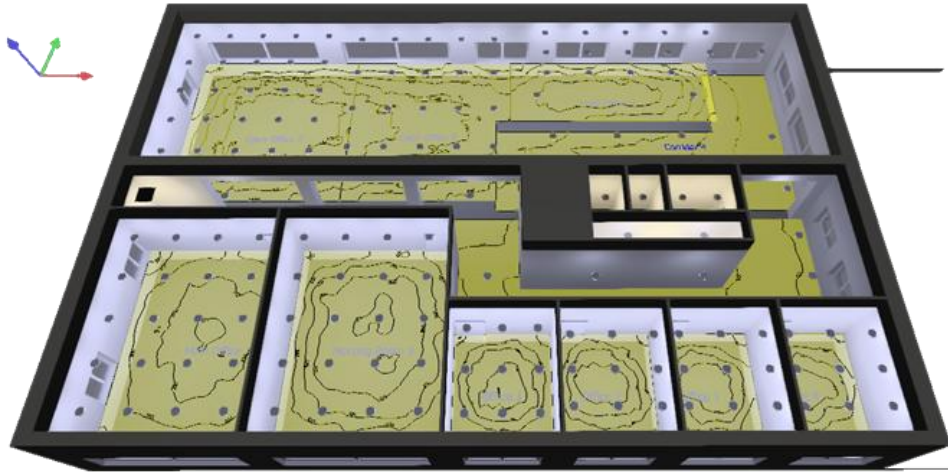
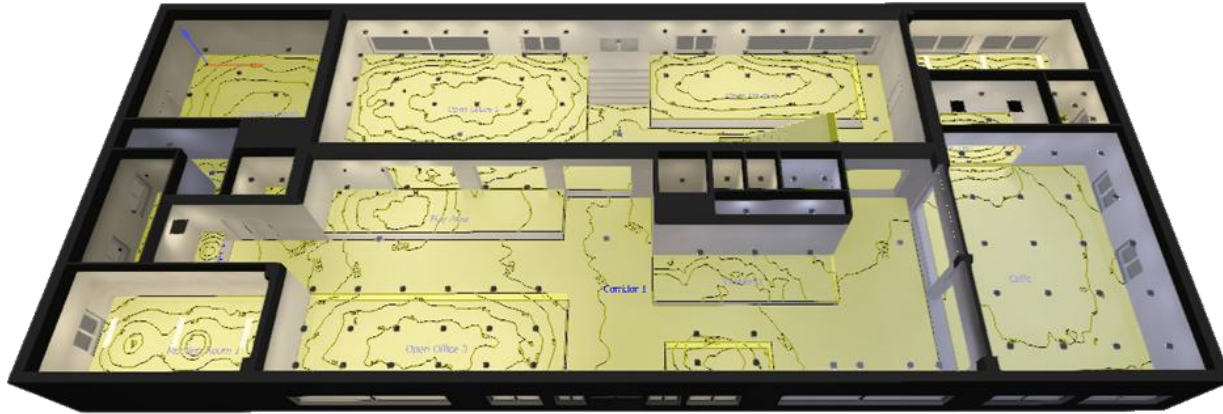
U-value	0.9
SHGC	0.605
T-vis	0.713



Old Windows

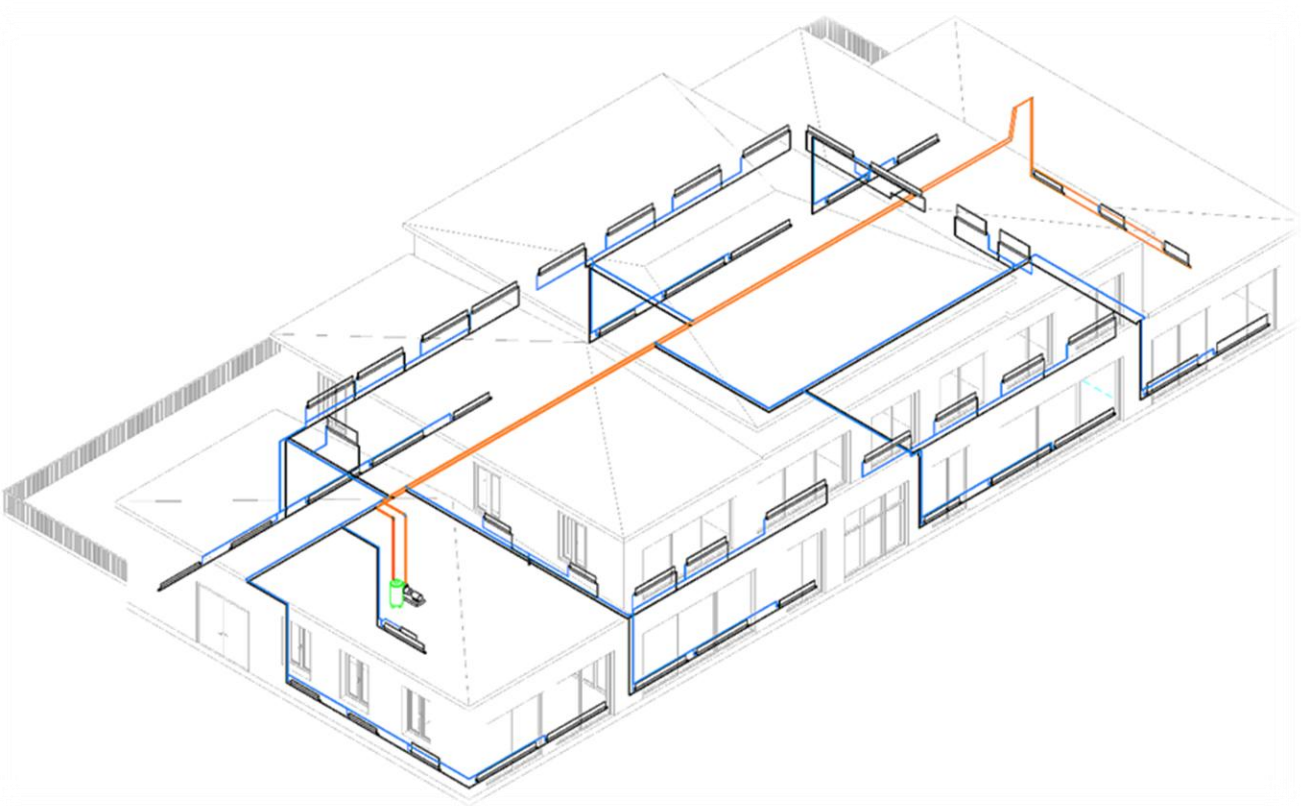
U-value	≈ 3.5
SHGC	≈ 0.6
T-vis	≈0.13-0.75
Mean	≈ 0.27

Electric Lighting



PARAMETER	VALUE	DETAILS
Maximum Energy Demand	5772 kWh/a	Peak annual energy consumption for lighting.
Estimated Energy Demand	4944 – 5772 kWh/a	Expected annual energy range, reflecting operational variability.
Maximum Energy Saving	828 kWh/a (14%)	Potential savings achievable through efficiency measures.
LENI	5 kWh/(m ² * a)	Lighting Energy Numeric Indicator, indicating overall energy efficiency.

Heating System



Total Heating Load – Ground Floor

13.42 kW

Enclosed Rooms Load – First Floor

6.96 kW

Total Heating Load (after 20% safety)	20.38 kW (13.42 kW Ground + 6.96 kW First Floor)
	10% of upper floor load added to ground floor
Nominal Flow Rate	1.151 L/s
	1.2 m/s
Measured Pressure Drop – Supply	4 317 Pa
	4 789 Pa
Total System Resistance	9 106 Pa
	0.93 m
Friction Limit (for Pipe Sizing)	150 Pa/m
	Copper
Thermal Expansion Coefficient	$17 \times 10^{-6} / ^\circ\text{C}$
	Central riser, ceiling branches
Balancing Strategy	Load split by window area & room zoning



Visualisations



Something to think about?

- The Importance of the Integrated Design Process (IDP)
- Daylight Performance Challenges Due to Extensive Greenery
- Selective Tree Trimming – A Likely Consideration



The background is a detailed architectural rendering of a modern building with a facade of vertical wooden slats. The building is surrounded by lush greenery, including trees and flowering plants. A large, white, circular inset on the left contains the text "Thank you for your attention!". Another circular inset on the right shows the interior of the building, which appears to be a large, open space with a high ceiling and a wooden floor. A third circular inset at the bottom center shows the entrance of the building, with a person walking towards the glass doors. The sky is blue with scattered white clouds.

Thank you for
your attention!

Any Questions?